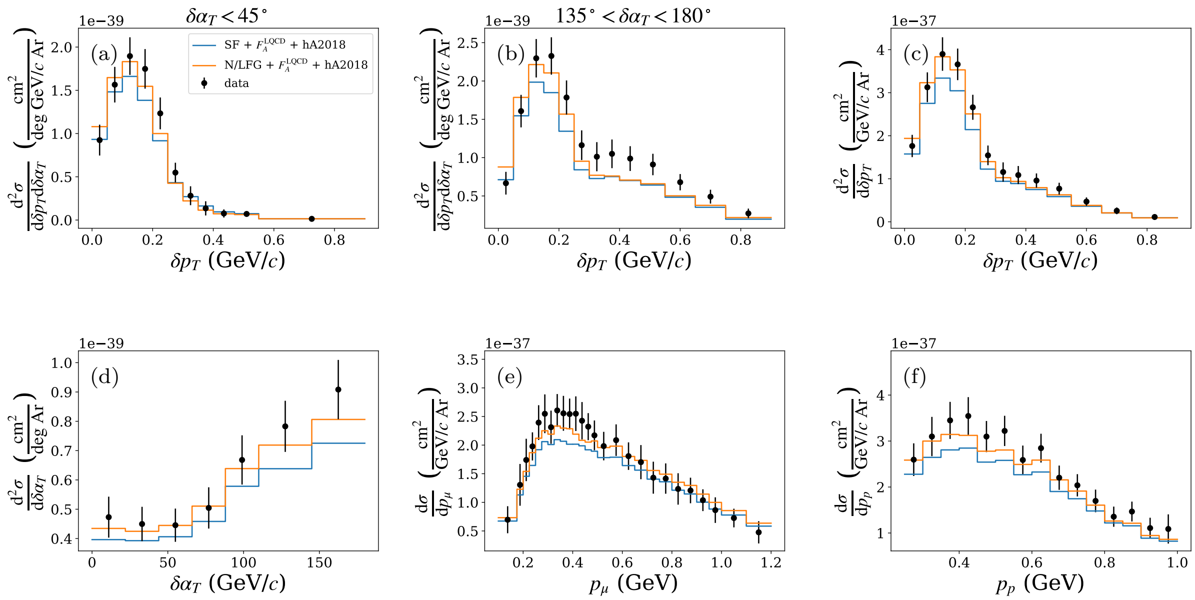

Figure 1: Effect of the nuclear ground-state model on the MicroBooNE CC1μ1p / CC1μNp argon cross sections with the LQCD axial form factor and hA2018 FSI held fixed: spectral function (blue) versus Local Fermi Gas (orange), compared with MicroBooNE data (black points). Panels: (a) δp_T for $\delta\alpha_T < 45^\circ$ and (b) for $135^\circ < \delta\alpha_T < 180^\circ$ (two slices of the 2D δp_T–$\delta\alpha_T$ measurement); (c) δp_T; (d) $\delta\alpha_T$; (e) p_μ; (f) p_p. With hA2018 the two ground states give very similar, good agreement with the data (small $\Delta\chi^2$/ndf in every observable; see Table 1).

Table 1: χ^2/ndf (p-value) relative to MicroBooNE data for Fig. 1 (LQCD axial form factor and hA2018 FSI held fixed).

Configuration	δp_T vs. $\delta\alpha_T$	δp_T	$\delta\alpha_T$	p_μ	p_p
SF + $F_A^{\rm LQCD}$ + hA2018	34.25/49 (0.95)	6.36/13 (0.93)	5.19/7 (0.64)	18.82/26 (0.84)	15.59/15 (0.41)
N/LFG + $F_A^{\rm LQCD}$ + hA2018	35.98/49 (0.92)	6.78/13 (0.91)	3.47/7 (0.84)	16.29/26 (0.93)	16.05/15 (0.38)

Table 2: χ^2/ndf (p-value) relative to MicroBooNE data for Fig. 2 (LQCD axial form factor and INCL FSI held fixed).

Configuration	δp_T vs. $\delta\alpha_T$	δp_T	$\delta\alpha_T$	p_μ	p_p
SF + $F_A^{\rm LQCD}$ + INCL	66.61/49 (0.05)	15.72/13 (0.26)	5.47/7 (0.60)	23.91/26 (0.58)	32.25/15 (0.01)
N/LFG + $F_A^{\rm LQCD}$ + INCL	56.01/49 (0.23)	9.09/13 (0.77)	5.75/7 (0.57)	17.66/26 (0.89)	26.94/15 (0.03)

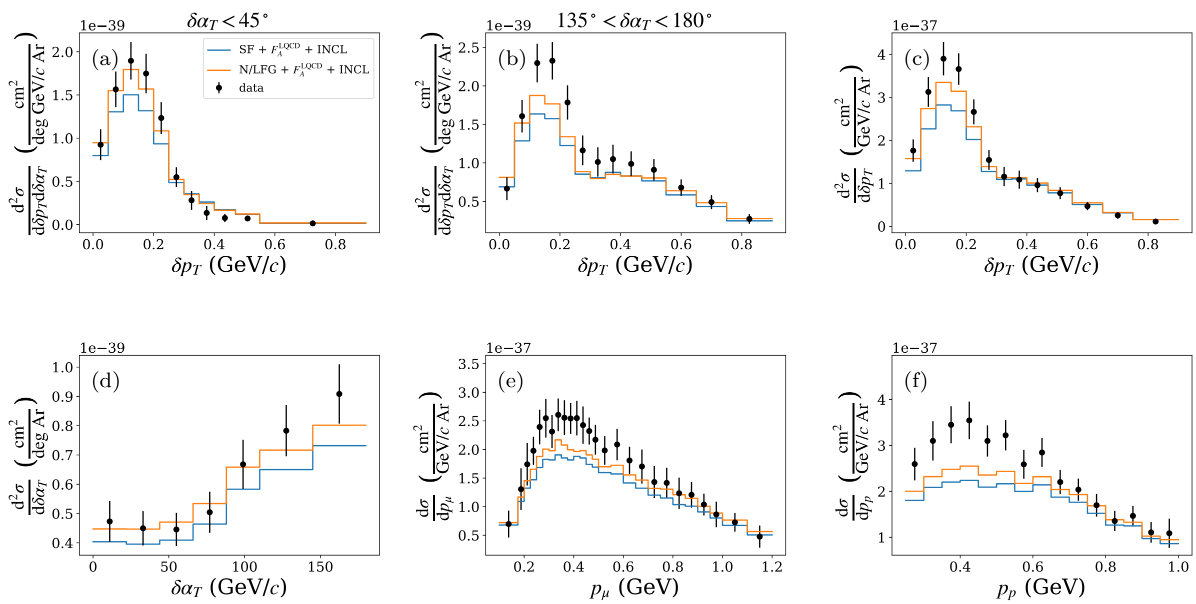

Figure 2: As in Fig. 1 but with INCL FSI (LQCD axial form factor held fixed): spectral function (blue) versus Local Fermi Gas (orange). With INCL the Local Fermi Gas agrees better than the spectral function, which raises χ^2/ndf in the 2D δp_T–$\delta\alpha_T$, δp_T and p_p (see Table 2).